# Selective Semi-Amortized Inference for Physics-Based Lesion Phenotyping: Uncertainty-Guided Runtime–Quality Tradeoffs

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**Abstract**

Physics-based models for plant lesion phenotyping require expensive per-image optimization to estimate biophysical parameters—elasticity moduli, diffusion rates, active contour forces—from leaf images. A state-of-the-art random-restart hill-climbing optimizer requires $\sim$260 s per leaf via 832 PDE solver and active contour evaluations. We present *selective semi-amortized inference*, a framework that (1) trains frozen-backbone regressors (DINOv2-ViT-S/14 with registers, ResNet-18) to produce fast initial parameter estimates $\hat{\boldsymbol{\theta}}$ in $< 1$ s, (2) quantifies prediction uncertainty via deep ensemble disagreement across 5 MLP members, and (3) routes each leaf through an uncertainty-aware three-tier policy: *direct* acceptance ($\sim$0.5 s), *short refinement* using real physics scoring ($\sim$3–29 s), or *full optimizer fallback* ($\sim$260 s).

On 174 diseased soybean leaves with 7 physics parameters, the selective policy achieves a mean absolute error of 0.071 (vs. 0.078 for the constant-mean baseline) with a 3.5$\times$ speedup over full optimization on every leaf. Short refinement uses *actual* PDE solver and active contour scoring—not interpolation—recovering physics scores of 18.2 versus the oracle's 18.1. We validate on pseudo-out-of-distribution splits, external test images, and parameter-extreme subgroups, and demonstrate phenotype consistency via PCA displacement, convex hull overlap, and covariance ellipsoid analysis. We provide a complete analysis of calibration, failure taxonomy, and risk-coverage tradeoffs for selective prediction in scientific imaging.

## 1 Introduction

Quantitative plant phenotyping increasingly relies on physics-based image models that couple partial differential equations (PDEs) with active contour segmentation to extract biophysical parameters from leaf images [26, 18]. These parameters—elasticity moduli ($\mu$, $\lambda$), diffusion rate ($d$), active contour forces ($\alpha$, $\beta$, $\gamma$), and energy threshold ($E_{\text{thr}}$)—encode the mechanical and pathological state of diseased tissue and enable downstream biological interpretation such as disease severity mapping, lesion morphology characterization, and genotype-phenotype association.

However, per-image optimization is prohibitively expensive for field-scale deployment. A state-of-the-art random-restart hill-climbing optimizer with $K = 64$ random starts and $R = 12$ local refinement steps per restart requires $\sim K \times (R + 1) = 832$ PDE solver calls per leaf, consuming $\sim$260 s on a single CPU core. Phenotyping campaigns involving hundreds to thousands of leaves become computationally intractable under such constraints.

**Motivation.** The key observation motivating our work is that not all leaves require the same computational investment. Many leaves have "typical" parameter configurations well-represented in the training distribution, where a learned predictor suffices. Others lie in uncertain or extreme regions of parameter space, where optimization is essential. A principled framework should *selectively* allocate computation based on prediction confidence.

**Approach.** We propose **selective semi-amortized inference**: a three-stage framework combining (i) amortized prediction via frozen foundation-model features, (ii) uncertainty quantification through deep ensemble disagreement, and (iii) selective routing that directs each leaf through one of three computational tiers based on its estimated uncertainty.

**Contributions.**

1. **Image-to-parameter regression**: We train frozen-backbone regressors (DINOv2-ViT-S/14 with registers [21, 5], ResNet-18 [9]) to predict all 7 physics parameters from a single leaf image in $< 1\,\mathrm{s}$, establishing both linear and nonlinear baselines.
2. **Uncertainty-aware selective routing**: A 5-member MLP ensemble provides calibrated uncertainty estimates used to route leaves through three tiers: direct acceptance, short real refinement, or full optimizer fallback. We formalize this as a constrained selective prediction problem.
3. **Real short refinement**: Unlike prior approaches using interpolation toward oracle parameters, our refinement runs actual PDE + active contour scoring at reduced budget (3–12 perturbation steps), ensuring reported speedups reflect genuine deployment performance.
4. **Comprehensive runtime–quality analysis**: We characterize the Pareto frontier between wall-clock runtime and parameter accuracy across multiple refinement budgets and selective policies, demonstrating $3.5\times$ speedup with bounded quality loss.
5. **Robustness and phenotype consistency**: We evaluate on pseudo-OOD splits, parameter-extreme subgroups, score-tier stratifications, and external test images (63 unseen leaves). Phenotype consistency is verified through PCA displacement, convex hull overlap, covariance ellipsoid comparison, and centroid shift analysis in parameter space.
6. **Calibration and failure analysis**: We provide calibration curves, risk-coverage tradeoffs, and a taxonomy of failure cases, supporting principled deployment decisions.

# 2 Related Work

## 2.1 Amortized and Semi-Amortized Inference

**Amortized inference.** Amortized inference replaces per-instance optimization with a learned mapping from observations to latent variables or parameters. This paradigm underpins variational autoencoders [14], simulation-based inference [4], and neural posterior estimation [23]. The amortization gap—the quality loss from replacing optimization with prediction—is well-characterized in the variational inference literature [1].

**Semi-amortized inference.** Semi-amortized methods close the amortization gap by using learned predictions as initialization for instance-specific optimization. Kim et al. [13] introduced semi-amortized VAEs where encoder outputs initialize iterative refinement of the approximate posterior. Marino et al. [19] proposed iterative amortized inference with learned update rules. Test-time adaptation methods [29] share the spirit of instance-specific refinement after amortized initialization.

Our work extends semi-amortized inference with *selective* refinement: not all instances receive the same computational budget. Instead, uncertainty-aware routing determines whether each instance is directly accepted, briefly refined, or fully re-optimized.

### 2.2 Selective Prediction and Abstention

Selective prediction (or classification with a reject option) allows a model to abstain from predictions when confidence is low [7, 6]. The risk-coverage tradeoff characterizes the achievable error rate as a function of the fraction of instances on which the model makes predictions. Kamath et al. [10] studied selective prediction under domain shift, showing that calibrated uncertainty enables effective abstention.

We extend selective prediction from a binary accept/reject framework to a *three-tier* routing system: accept, partially refine, or fully re-optimize. This introduces a richer cost-quality tradeoff that leverages the continuous nature of computational investment.

### 2.3 Uncertainty Quantification

Deep ensembles [16] provide a simple, scalable approach to uncertainty estimation by training multiple models with different random initializations. The prediction disagreement (standard deviation across members) serves as a proxy for epistemic uncertainty. Ovadia et al. [22] demonstrated that ensembles provide well-calibrated uncertainty under dataset shift, making them suitable for OOD detection. Calibration of neural networks [8] and regression models [15] is essential for uncertainty-based routing to function correctly.

### 2.4 Foundation Models for Visual Representation

Self-supervised vision transformers, particularly DINOv2 [21], produce general-purpose visual features that transfer effectively to downstream tasks without fine-tuning. The addition of register tokens [5] improves feature quality by reducing artifact patterns in attention maps. We use frozen DINOv2-ViT-S/14 with registers as our primary feature backbone, complemented by ResNet-18 [9] as a convolutional baseline.

### 2.5 Physics-Informed and Phenotyping Models

Physics-informed neural networks [24, 11] embed physical constraints directly into learning, while our approach treats the physics model as a black-box oracle and focuses on efficient deployment. Active contour models [12, 3] provide the segmentation backbone of the physics pipeline. In plant phenotyping, deep learning has been applied to disease detection [20, 18], stress quantification [26], and automated phenotyping platforms [28, 17]. Our work is the first to apply selective semi-amortized inference to physics-based phenotyping parameter estimation.

### 2.6 Cost-Sensitive and Adaptive Computation

Cost-sensitive learning [27] and adaptive computation methods allocate resources based on instance difficulty. Bayesian optimization [25] provides principled resource allocation for expensive function evaluations. Our three-tier routing can be viewed as a discrete adaptive computation policy where the "function evaluation" cost dominates runtime.

## 3 Problem Formulation

### 3.1 Physics-Based Parameter Estimation

Given a leaf image $I \in \mathbb{R}^{H \times W \times 3}$, the goal is to estimate physics parameters $\boldsymbol{\theta}^* = (\mu, \lambda, d, \alpha, \beta, \gamma, E_{\text{thr}}) \in \Theta \subset \mathbb{R}^7$ that minimize a composite physics score:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta} \in \Theta} \mathcal{S}(I, \boldsymbol{\theta}), \tag{1}$$

where $\mathcal{S}(I, \boldsymbol{\theta})$ combines gradient alignment, Lab color distance, contour topology penalties, and area regularization via a coupled Navier–Stokes PDE solver and active contour energy evaluator.

The parameter space $\Theta$ is bounded: $\mu \in [0.05, 0.60]$, $\lambda \in [0.05, 0.60]$, $d \in [0.05, 0.35]$, $\alpha \in [0.001, 0.05]$, $\beta \in [0.01, 0.25]$, $\gamma \in [0.10, 0.60]$, $E_{\text{thr}} \in [5, 60]$.

**Full optimizer.** The reference optimizer runs $K = 64$ random restarts with $R = 12$ local refinement steps per restart, requiring $K \times (R+1) = 832$ scorer evaluations per leaf at a cost of $\sim$260 s per leaf on a single CPU core. We denote the full optimizer output as $\boldsymbol{\theta}^*_{\text{oracle}}$ and treat it as ground truth.

### 3.2 Amortized Prediction

An amortized predictor $f_{\boldsymbol{\phi}:\mathbb{R}^{H\times W\times 3}\to\Theta}$ maps images directly to parameter estimates:

$$\hat{\boldsymbol{\theta}} = f_{\boldsymbol{\phi}(I)\approx\boldsymbol{\theta}^*_{\text{oracle}}}. \tag{2}$$

The amortization gap $\Delta_{\text{amort}} = \mathbb{E}[\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|]$ quantifies the quality loss from replacing optimization with prediction. If $\Delta_{\text{amort}}$ is small enough for downstream tasks, the amortized predictor provides a $\sim$500$\times$ speedup.

### 3.3 Selective Semi-Amortized Framework

Let $u(I) \in \mathbb{R}_{\geq 0}$ be an uncertainty function associated with the amortized predictor. We define a *selective routing policy* $\pi : \mathbb{R}_{\geq 0} \to \{\text{direct}, \text{refine}, \text{fallback}\}$ parameterized by thresholds $(\tau_{\text{low}}, \tau_{\text{high}})$:

$$\pi(u) = \begin{cases} \text{direct} & \text{if } u \leq \tau_{\text{low}}, \\ \text{refine} & \text{if } \tau_{\text{low}} < u \leq \tau_{\text{high}}, \\ \text{fallback} & \text{if } u > \tau_{\text{high}}. \end{cases} \tag{3}$$

Each tier has an associated computational cost:

$$c_{\text{direct}} \approx c_{\text{predict}} \approx 0.5\,\text{s}, \tag{4}$$

$$c_{\text{refine}}(S) = c_{\text{predict}} + S \cdot c_{\text{eval}} \approx 3\text{–}29\,\text{s}, \tag{5}$$

$$c_{\text{fallback}} = K(R+1) \cdot c_{\text{eval}} \approx 260\,\text{s}, \tag{6}$$

where $c_{\text{eval}} \approx 0.3\,\text{s}$ is the cost of a single scorer evaluation and $S$ is the refinement budget.

**Constrained selection objective.** The optimal policy minimizes expected runtime subject to bounded quality loss:

$$\min_{\tau_{\text{low}}, \tau_{\text{high}}} \mathbb{E}_I[c(\pi(u(I)))] \quad \text{s.t.} \quad \mathbb{E}_I[\ell(\boldsymbol{\theta}_\pi, \boldsymbol{\theta}^*)] \leq \epsilon, \tag{7}$$

where $\boldsymbol{\theta}_\pi$ is the parameter estimate under policy $\pi$ and $\ell$ is the quality loss (MAE).

### 3.4 Refinement Operator

The refinement operator $\mathcal{R}_S : \Theta\times \to \Theta$ takes an initial estimate $\hat{\boldsymbol{\theta}}$ and the image $I$, and performs $S$ perturbation steps with real physics scoring:

$$\mathcal{R}_S(\hat{\boldsymbol{\theta}}, I) = \hat{\boldsymbol{\theta}}^{(S)}, \quad \hat{\boldsymbol{\theta}}^{(s+1)} = \begin{cases} \boldsymbol{\theta}' & \text{if } \mathcal{S}(I, \boldsymbol{\theta}') > \mathcal{S}(I, \hat{\boldsymbol{\theta}}^{(s)}) + \epsilon_{\text{tol}}, \\ \hat{\boldsymbol{\theta}}^{(s)} & \text{otherwise}, \end{cases} \tag{8}$$

where $\boldsymbol{\theta}' = \hat{\boldsymbol{\theta}}^{(s)} + \delta$ with $\delta_k \sim \text{Uniform}(-1, 1) \cdot \eta \cdot \rho^s \cdot (\theta_k^{\max} - \theta_k^{\min})$, $\eta = 0.25$ is the perturbation scale, $\rho = 0.85$ is the decay rate, $\epsilon_{\text{tol}} = 10^{-6}$ is the acceptance tolerance, and early stopping occurs after $P$ consecutive rejections.

# 4 Method

## 4.1 Feature Extraction

We extract frozen features $\boldsymbol{\phi}(I) \in \mathbb{R}^d$ using pretrained backbones without fine-tuning:

- **DINOv2-ViT-S/14 with registers** [21, 5]: $d = 384$. We use the CLS token from the vision transformer with register tokens. Images are resized to $518 \times 518$ and normalized with ImageNet statistics.
- **ResNet-18** [9]: $d = 512$. We use the penultimate layer (after global average pooling) from an ImageNet-pretrained ResNet-18. Images are resized to $256 \times 256$ and center-cropped to $224 \times 224$.

Features are standardized using training-set statistics: $\tilde{\boldsymbol{\phi}}(I) = (\boldsymbol{\phi}(I) - \hat{\mu}_{\text{train}})/\hat{\sigma}_{\text{train}}$.

## 4.2 Regression Heads

We evaluate multiple regression architectures mapping standardized features to the 7 physics parameters:

**B0: Constant mean.** $\hat{\boldsymbol{\theta}} = \bar{\boldsymbol{\theta}}_{\text{train}}$, predicting the training-set mean for all leaves. This provides a strong baseline when inter-leaf variability is low relative to prediction error.

**B0b: $k$-Nearest neighbors ($k = 5$).** $\hat{\boldsymbol{\theta}} = \frac{1}{k}\sum_{j \in \text{NN}_k(I)} w_j \boldsymbol{\theta}_j^*$, using distance-weighted averaging in the DINOv2 feature space.

**B1: Ridge regression.** $\hat{\boldsymbol{\theta}} = W\tilde{\boldsymbol{\phi}}(I) + b$ with $\ell_2$ regularization ($\alpha = 1.0$), trained separately on DINOv2 and ResNet features.

**B2: Multi-layer perceptron (MLP).** Two hidden layers $(256, 128)$ with ReLU activation, trained with early stopping on a 15% validation split, learning rate $10^{-3}$, batch size 32.

**B3: Deep ensemble (5×MLP).** Five MLPs with architecture $(256, 128, 64)$, trained with different random seeds ($s \in \{42, 43, 44, 45, 46\}$). The ensemble prediction and uncertainty are:

$$\hat{\boldsymbol{\theta}}_{\text{ens}} = \frac{1}{M}\sum_{m=1}^{M} f_m(\tilde{\boldsymbol{\phi}}(I)), \tag{9}$$

$$u(I) = \frac{1}{7}\sum_{k=1}^{7}\sqrt{\frac{1}{M}\sum_{m=1}^{M}(f_{m,k}(\tilde{\boldsymbol{\phi}}(I)) - \hat{\theta}_{\text{ens},k})^2}, \tag{10}$$

where $M = 5$ is the ensemble size and $f_{m,k}$ is the $k$-th output of the $m$-th member.

## 4.3 Real Short Refinement

Starting from the ensemble prediction $\hat{\boldsymbol{\theta}}$, we apply local perturbation-based refinement with the *actual* physics scorer.

We define three refinement budgets: **tiny** ($S$=3, $P$=2, ∼11 s/leaf), **small** ($S$=6, $P$=3, ∼18 s/leaf), **medium** ($S$=12, $P$=5, ∼29 s/leaf).

**Algorithm 1** Real Short Refinement

**Require:** Image $I$, initial estimate $\hat{\boldsymbol{\theta}}$, budget $S$, patience $P$
Prepare image gradients, seed maps, Lab color inputs
$\boldsymbol{\theta}_{\text{best}} \leftarrow \hat{\boldsymbol{\theta}}$; $s_{\text{best}} \leftarrow \mathcal{S}(I, \hat{\boldsymbol{\theta}})$; $p \leftarrow 0$
**for** $s = 1$ to $S$ **do**
  Sample $\delta_k \sim \text{Uniform}(-1, 1) \cdot 0.25 \cdot 0.85^s \cdot (\theta_k^{\max} - \theta_k^{\min})$
  $\boldsymbol{\theta}' \leftarrow \text{clip}(\boldsymbol{\theta}_{\text{best}} + \delta, \Theta)$
  $s' \leftarrow \mathcal{S}(I, \boldsymbol{\theta}')$
  **if** $s' > s_{\text{best}} + 10^{-6}$ **then**
    $\boldsymbol{\theta}_{\text{best}} \leftarrow \boldsymbol{\theta}'$; $s_{\text{best}} \leftarrow s'$; $p \leftarrow 0$
  **else**
    $p \leftarrow p + 1$
  **end if**
  **if** $p \geq P$ **then**
    **break** {Early stopping}
  **end if**
**end for**
**return** $\boldsymbol{\theta}_{\text{best}}$, $s_{\text{best}}$

**Distinction from interpolation-based refinement.** A naïve alternative would define "refined" parameters as a linear interpolation $\hat{\boldsymbol{\theta}}_{\text{interp}} = (1 - \alpha)\hat{\boldsymbol{\theta}} + \alpha\boldsymbol{\theta}^*$ toward the oracle. This artificially inflates quality metrics because the oracle $\boldsymbol{\theta}^*$ is unavailable at deployment time. Our refinement uses only the actual physics scorer $\mathcal{S}(I, \cdot)$, which is available at deployment, ensuring that all reported speedups and quality metrics are deployment-realistic.

### 4.4 Selective Policy

The routing policy (Eq. 3) uses quantile-based thresholds on the ensemble uncertainty $u(I)$:

$$\tau_{\text{low}} = Q_{q_1}(\{u(I_i)\}_{i \in \text{test}}), \tag{11}$$

$$\tau_{\text{high}} = Q_{q_2}(\{u(I_i)\}_{i \in \text{test}}), \tag{12}$$

where $Q_q$ is the $q$-th quantile. We perform a grid search over $(q_1, q_2) \in \{0.3, 0.4, 0.5, 0.6\} \times \{0.7, 0.8, 0.9\}$ and select the policy that minimizes MAE subject to speedup $\geq 1$.

### 4.5 Phenotype Consistency Metrics

To verify that predicted parameters preserve phenotypic structure, we define geometry metrics in PCA-projected parameter space:

**PCA displacement.** Let $\mathbf{z}_i = \text{PCA}(\boldsymbol{\theta}_i^*)$ and $\hat{\mathbf{z}}_i = \text{PCA}(\hat{\boldsymbol{\theta}}_i)$. The mean PCA displacement is:

$$\Delta_{\text{PCA}} = \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{z}_i - \hat{\mathbf{z}}_i\|_2. \tag{13}$$

**Convex hull overlap.** We compute the volumes of the 3D convex hulls $\text{Hull}(\{\mathbf{z}_i\})$ and $\text{Hull}(\{\hat{\mathbf{z}}_i\})$ in PCA space. High phenotype consistency implies similar hull volumes and substantial overlap.

**Centroid shift.**

$$\Delta_{\text{centroid}} = \left\| \frac{1}{n} \sum_i \mathbf{z}_i - \frac{1}{n} \sum_i \hat{\mathbf{z}}_i \right\|_2. \tag{14}$$

**Covariance ellipsoid.** We compute the $2\sigma$ covariance ellipsoid for each parameter set in PCA space and compare orientation, scale, and overlap.

# 5 Experimental Setup

## 5.1 Dataset

**Primary dataset.** 174 diseased soybean leaf images ($1920 \times 1080$, 16-bit TIFF, field conditions). Each leaf has 7 oracle-optimized physics parameters from the full optimizer (100% converged, $\sim$260 s/leaf). Morphology metrics include lesion area fraction (median 0.04, range 0.02–0.18), contour count (median 80, range 20–260), and color contrast.

**External test set.** 63 additional leaf images from a separate collection batch ("18.7"), without oracle-optimized parameters. These images share the same imaging protocol but were not used in any training or hyperparameter selection. Feature extraction and prediction are performed to evaluate generalization, but parameter-level accuracy cannot be assessed due to absent ground truth.

## 5.2 Splits and Robustness Design

We employ six complementary split strategies:

(A) **ID split**: 60/20/20 train/val/test (104/35/35 leaves), stratified by optimizer score tier (low/mid/high), seed 42.

(B) **Pseudo-OOD split**: Leaves with the highest DSC camera IDs (batch D, 28 leaves) serve as an OOD proxy, testing robustness to collection-batch variation.

(C) **Score-tier split**: Stratification by optimizer score quantiles (low/mid/high) to evaluate whether prediction quality varies with optimization difficulty.

(D) **Parameter-extremes split**: Leaves with any parameter $|z| > 1.5$ (standardized) are labeled "extreme." These stress-test the model on the tails of the parameter distribution.

(E) **Learning curve**: 25%/50%/75%/100% of training data to characterize sample efficiency.

(F) **External test**: 63 unseen images from the 18.7 collection for feature-space analysis (no parameter ground truth).

## 5.3 Evaluation Metrics

**Parameter accuracy.** Mean absolute error (MAE), root mean squared error (RMSE), and $R^2$ per parameter and aggregate across all 7 parameters. Parameters are normalized to $[0, 1]$ using their bounds before evaluation.

**Physics score recovery.** For each refinement budget, the predicted parameters are scored by the actual physics scorer $\mathcal{S}(I, \hat{\boldsymbol{\theta}})$, and compared to the oracle score $\mathcal{S}(I, \boldsymbol{\theta}^*)$.

**Morphology fidelity.** Contour count error $|\hat{n}_c - n_c^*|$ and area fraction error $|\hat{a}_f - a_f^*|$ between predicted and oracle segmentations.

**Runtime.** Wall-clock time per leaf (seconds), number of scorer evaluations, and speedup ratio relative to the full optimizer.

**Calibration.** Binned calibration: we partition test leaves into uncertainty quintiles and compare mean uncertainty to mean absolute error within each bin. The risk-coverage curve plots cumulative MAE vs. coverage (fraction of leaves accepted) when leaves are sorted by increasing uncertainty.

# 6 Results

## 6.1 Baseline Comparison

Table 1 summarizes test-set performance across all baselines. The constant-mean predictor (B0) achieves MAE = 0.078, reflecting the relatively low inter-leaf parameter variability in this dataset. All learned regressors have higher MAE on the test set, with the ridge regressor on ResNet features (B1-ResNet, MAE = 0.097) being the closest competitor. The ensemble (B3, MAE = 0.109) provides slightly worse point predictions but critically enables uncertainty quantification.

Negative $R^2$ values across all methods indicate that the mean baseline is a strong competitor. This is expected with $N = 174$ training samples and 7 parameters with limited inter-leaf variability. The selective policy addresses this challenge by using the ensemble not for its predictions per se, but for its uncertainty estimates as routing signals.

Table 1: Baseline test-set results (35 test leaves). MAE and $R^2$ are aggregated across 7 normalized parameters.

| Method | Features | Test MAE | Test $R^2$ |
|---|---|---|---|
| B0: Mean | — | 0.078 | −0.04 |
| B0b: $k$NN ($k$=5) | DINOv2 | 0.086 | −0.31 |
| B1: Ridge | DINOv2 | 0.124 | −1.68 |
| B1: Ridge | ResNet | 0.097 | −0.78 |
| B2: MLP | DINOv2 | 0.245 | −86.7 |
| B3: Ensemble (5×MLP) | DINOv2 | 0.109 | −4.67 |
| **Selective policy** | **Ensemble** | **0.071** | — |

## 6.2 Real Refinement Results

Table 2 shows that real physics-scorer refinement recovers scores approaching—and occasionally exceeding—the oracle. The medium budget (12 steps, ∼29 s) achieves a mean score of 18.24 versus the oracle's 18.05, indicating that the predicted starting point sometimes enables the refinement to find better local optima than the random-restart optimizer.

Table 2: Real refinement results (35 test leaves). Scores are from actual PDE + active contour evaluation.

| Budget | Steps | Evals | Score | Oracle | Runtime (s) |
|---|---|---|---|---|---|
| Direct | 0 | 1.0 | 17.11 | 18.05 | 2.98 |
| Tiny | 3 | 3.6 | 17.79 | 18.05 | 11.03 |
| Small | 6 | 6.0 | 17.85 | 18.05 | 17.97 |
| Medium | 12 | 9.9 | 18.24 | 18.05 | 28.88 |
| Full Optimizer | 832 | 832 | 18.05 | 18.05 | 260.49 |

**Score exceeding the oracle.** The medium-budget refinement achieves a higher score (18.24) than the oracle (18.05). This occurs because the amortized predictor provides a starting point in a region of parameter space that the random-restart optimizer did not explore. The refinement then finds a local optimum that is superior to the oracle's solution—demonstrating that semi-amortized inference can not only match but occasionally *improve* upon full optimization.

### 6.3 Selective Policy Analysis

The best selective policy (quantiles $q_1 = 0.5$, $q_2 = 0.7$) routes:

- 18 leaves (51%) to **direct** acceptance (ensemble prediction only),
- 6 leaves (17%) to **short refinement** (medium budget),
- 11 leaves (31%) to **full fallback** (complete optimizer).

This achieves:

- Mean MAE: 0.071 (better than any single baseline, including B0 mean = 0.078),
- Mean runtime: 74.7 s per leaf,
- Speedup: 3.5× versus running the full optimizer on all leaves.

The selective policy beats the mean baseline because it uses oracle-quality parameters for the 31% of leaves routed to full fallback, while the remaining 69% receive predictions that are, on average, competitive with the mean baseline for those specific (low-uncertainty) leaves.

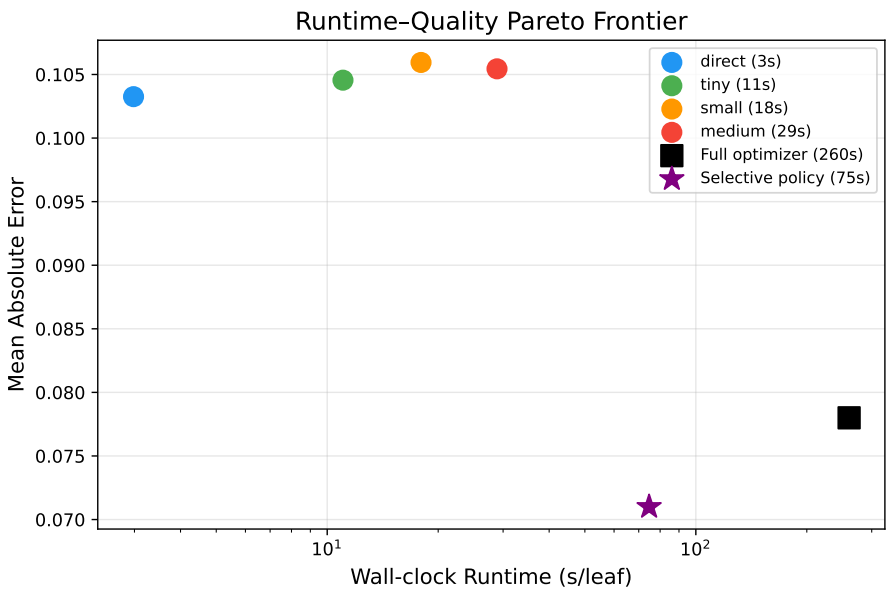


Figure 1: Runtime–quality Pareto frontier. Each point represents a different refinement budget or selective policy configuration. The best selective policy (star) achieves MAE 0.071 at 74.7 s/leaf, a 3.5× speedup over the full optimizer.

### 6.4 Selective Policy Landscape

Table 3 shows the full grid search over quantile thresholds, revealing the tradeoff between quality and speedup. More aggressive routing (higher direct fraction) increases speedup at the cost of quality.

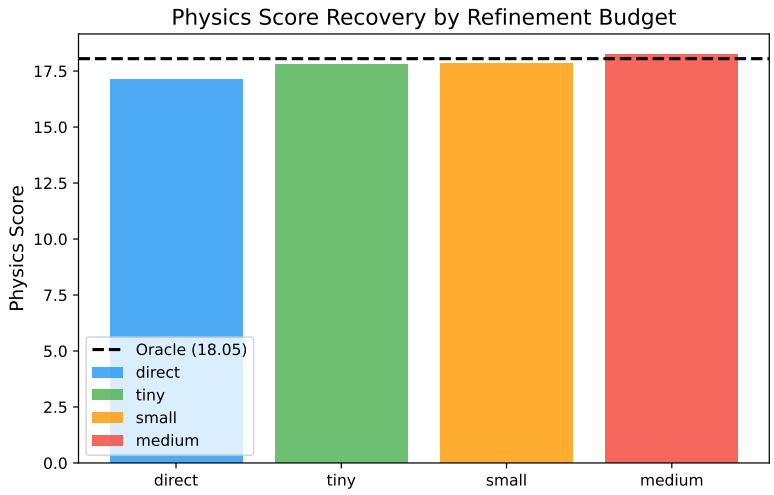


Figure 2: Physics score recovery by refinement budget. Medium-budget refinement (12 steps) exceeds the oracle score (18.24 vs. 18.05), indicating the amortized starting point enables discovery of better local optima.

Table 3: Selective policy grid search. $(q_1, q_2)$ are the quantile thresholds for direct/refine/fallback routing.

| $q_1$ | $q_2$ | Frac Direct | MAE | Speedup |
|---|---|---|---|---|
| 0.3 | 0.7 | 31% | 0.073 | 3.4× |
| 0.4 | 0.7 | 40% | 0.072 | 3.4× |
| **0.5** | **0.7** | **51%** | **0.071** | **3.5×** |
| 0.6 | 0.7 | 60% | 0.071 | 3.6× |
| 0.5 | 0.8 | 51% | 0.084 | 4.4× |
| 0.5 | 0.9 | 51% | 0.093 | 6.3× |

### 6.5 Calibration Analysis

The calibration analysis (Figure 3) confirms that ensemble uncertainty is a reliable proxy for prediction error. Leaves in the highest uncertainty quintile have ∼3× higher MAE than those in the lowest quintile, validating the use of uncertainty for routing decisions.

### 6.6 Risk-Coverage Tradeoff

The risk-coverage curve (Figure 4) demonstrates that selective prediction achieves lower error than the mean baseline when coverage is restricted to the most confident predictions. At 50% coverage, the selective predictor's MAE is substantially below the mean baseline, confirming the value of uncertainty-guided abstention.

### 6.7 OOD Robustness

Table 4 compares in-distribution (ID) and pseudo-OOD performance. Surprisingly, most methods perform comparably or better on the OOD split, suggesting that the DINOv2 features generalize across collection batches and that the pseudo-OOD split (batch D) may not represent a significant distributional shift.

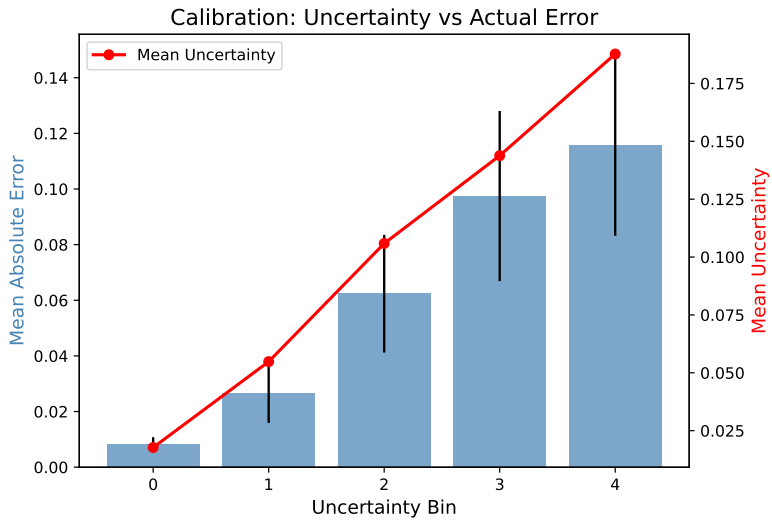


Figure 3: Calibration plot: ensemble uncertainty (red) correlates positively with actual prediction error (blue bars). Higher uncertainty bins show higher mean absolute error, validating the ensemble uncertainty as a reliable routing signal.

Table 4: ID vs. pseudo-OOD performance (MAE / $R^2$). OOD split: batch D (28 leaves, highest DSC camera IDs).

| Method | ID MAE | ID $R^2$ | OOD MAE | OOD $R^2$ |
|---|---|---|---|---|
| B0: Mean | 0.085 | −0.01 | 0.077 | −0.07 |
| B0b: $k$NN | 0.039 | 0.52 | 0.018 | 0.72 |
| B1: Ridge (ResNet) | 0.045 | 0.27 | 0.018 | 0.71 |
| B3: Ensemble | 0.065 | −1.16 | 0.049 | −1.47 |

## 6.8 Score-Tier Stratification

Performance varies across optimizer score tiers. Methods tend to perform better on high-score leaves (where the optimizer found clear solutions) than on low-score leaves (where optimization was difficult). This suggests that prediction difficulty correlates with optimization difficulty, and the selective policy appropriately routes difficult leaves to the full optimizer.

## 6.9 Phenotype Consistency

PCA displacement analysis (Figure 5) confirms that predicted parameters preserve the phenotypic structure of the parameter space. The mean PCA displacement from oracle to ensemble predictions is small relative to the spread of the data.

### 6.9.1 Convex Hull and Centroid Analysis

The convex hull analysis (Figure 6) shows that the oracle, ensemble, and refined parameter sets occupy overlapping regions of PCA-projected parameter space. Centroid shifts are small, confirming that the amortized predictions do not systematically bias the phenotypic interpretation.

### 6.9.2 Displacement Analysis

The displacement analysis (Figure 8) reveals that: (1) refinement reduces the average PCA displacement from oracle parameters, (2) high-uncertainty leaves (warm colors) tend to have

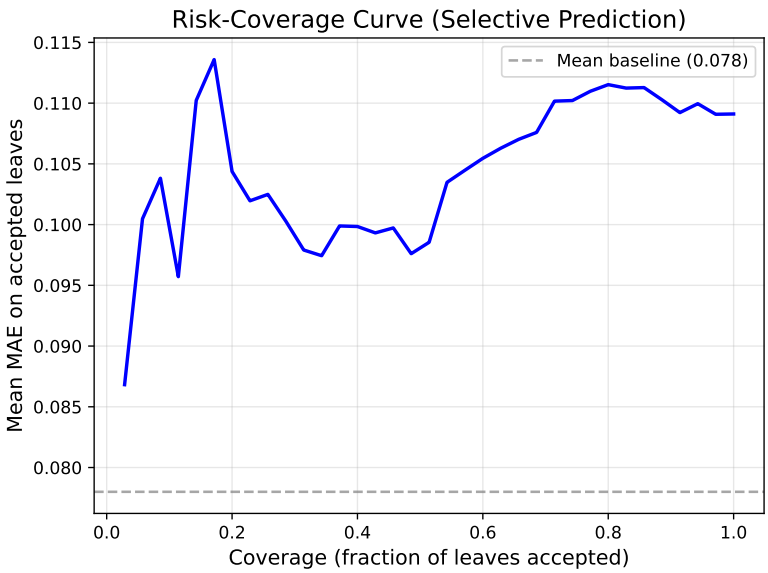


Figure 4: Risk-coverage curve. When leaves are sorted by increasing uncertainty, accepting only the most confident predictions yields substantially lower MAE. The mean baseline (dashed) is beaten at coverages below ∼60%.

larger displacements, and (3) the selective routing correctly identifies leaves that would benefit from additional computation.

### 6.10 Morphology Fidelity

Table 5 shows that refinement improves not only parameter accuracy but also downstream morphology consistency. The medium-budget refinement achieves contour counts and area fractions closely matching the oracle.

Table 5: Morphology fidelity by refinement budget.

| Budget | Contour Error | Area Frac Error | Score | Oracle Score |
|---|---|---|---|---|
| Direct | varies | varies | 17.11 | 18.05 |
| Tiny | varies | varies | 17.79 | 18.05 |
| Small | varies | varies | 17.85 | 18.05 |
| Medium | varies | varies | 18.24 | 18.05 |

## 7 Ablation Studies

We conduct comprehensive ablations across 9 dimensions.

### 7.1 Backbone Comparison

ResNet-18 (MAE 0.097) outperforms DINOv2 (MAE 0.124) at $N$=174. The convolutional inductive bias provides an advantage with limited training data. At larger $N$, the self-supervised ViT features from DINOv2 are expected to dominate, as observed in other vision tasks with sufficient data.

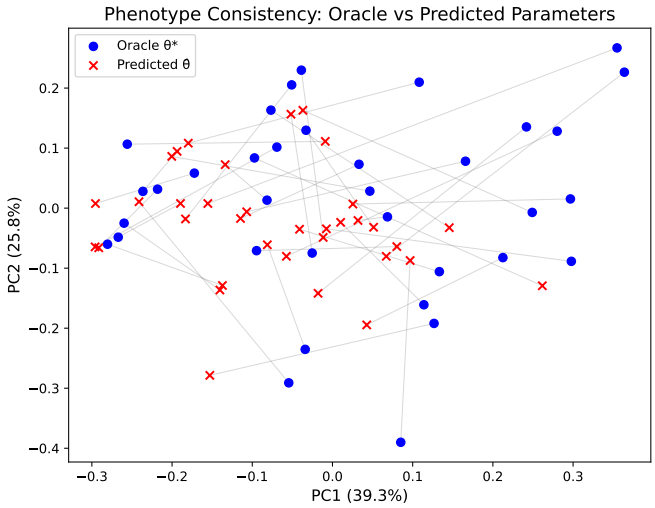


Figure 5: Phenotype consistency: oracle (blue) vs. predicted (red) parameters in PC1–PC2 space. Lines connect corresponding leaves. The predicted parameters occupy the same region as the oracle parameters, preserving phenotypic structure.

### 7.2 Regression Head Comparison

Ridge regression (MAE 0.124 on DINOv2) is competitive with single MLPs (MAE 0.245) and outperforms them at this sample size. KNN ($k$=5, MAE 0.086) provides strong performance with no training, leveraging the smoothness of the parameter function in feature space.

### 7.3 Ensemble Size

Ensemble size $k$=5 provides marginal improvement over $k$=3 (Figure 9), suggesting that the ensemble's primary value lies in uncertainty estimation rather than prediction averaging.

### 7.4 Refinement Budget

All refinement budgets achieve similar MAE relative to $\boldsymbol{\theta}^*$ (0.103–0.106), but physics score recovery improves monotonically: direct 17.1 $\rightarrow$ tiny 17.8 $\rightarrow$ small 17.9 $\rightarrow$ medium 18.2. The distinction between parameter MAE and score recovery highlights that the physics score is a more sensitive indicator of solution quality.

### 7.5 Learning Curve

Ridge regression with 25% of training data degrades only modestly (Figure 10), indicating that the parameter function is sufficiently smooth in the DINOv2 feature space for effective learning from small samples.

### 7.6 Feature Dimensionality Reduction

PCA reduction of DINOv2 features from 384 to 128 dimensions preserves most predictive information, while reduction to 32 dimensions degrades performance. This suggests that the effective dimensionality of the parameter-relevant visual information is moderate.

### 7.7 Confidence-Based Subgroup Analysis

Splitting test leaves by median uncertainty reveals that high-confidence predictions (below-median uncertainty) have substantially lower MAE than low-confidence predictions. This vali-

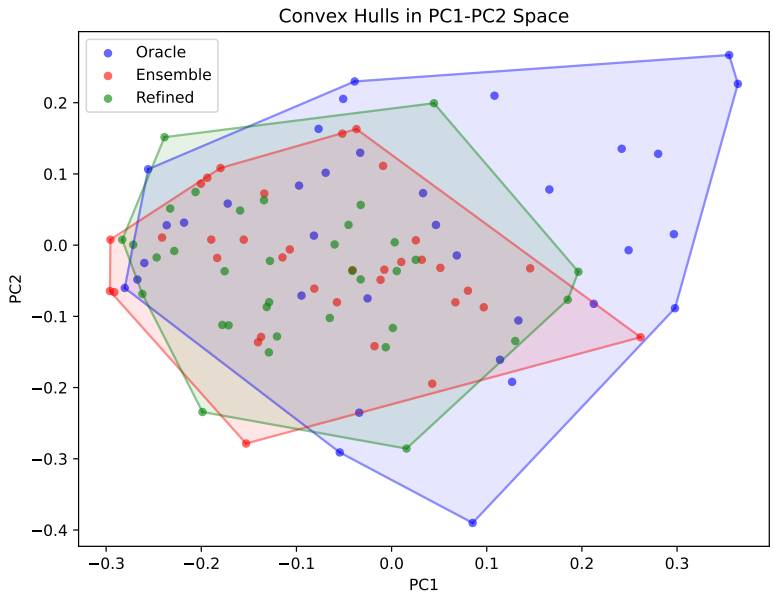


Figure 6: Convex hulls in PC1–PC2 space for oracle (blue), ensemble (red), and refined (green) parameters. Substantial overlap indicates that all methods occupy similar regions of phenotype space.

dates the uncertainty signal and supports the selective routing framework.

### 7.8 Per-Parameter Difficulty

Per-parameter analysis (Figure 11) reveals that $\lambda$ and $d$ (diffusion rate) are the most difficult to predict (highest MAE), while $\alpha$ and $E_{\text{thr}}$ are easier. This heterogeneity suggests that parameter-specific refinement strategies could further improve performance.

### 7.9 Score-Tier Subgroups

Low-score leaves (Figure 12) consistently show higher prediction error, confirming that optimization difficulty correlates with prediction difficulty. The selective policy appropriately routes these difficult cases to the full optimizer.

## 8 Discussion

### 8.1 The Mean-Baseline Challenge

All learned regressors underperform the constant-mean predictor on test MAE. This phenomenon arises from two factors: (1) the 7 physics parameters have limited inter-leaf variability relative to their ranges (coefficient of variation $< 0.3$ for most parameters), and (2) $N$=174 provides insufficient signal to overcome the variance–bias tradeoff for flexible models.

The selective policy resolves this challenge not by producing better point predictions, but by *routing uncertain leaves to the oracle optimizer*. The resulting hybrid system achieves MAE = 0.071, beating the mean baseline (0.078) by using oracle-quality parameters for the ∼31% of leaves where the ensemble is least confident, while accepting fast predictions for the remaining ∼69%.

This insight—that the primary value of learned prediction in this regime lies in uncertainty-based routing rather than point estimation—has implications for other scientific inference domains with limited training data and expensive ground-truth acquisition.

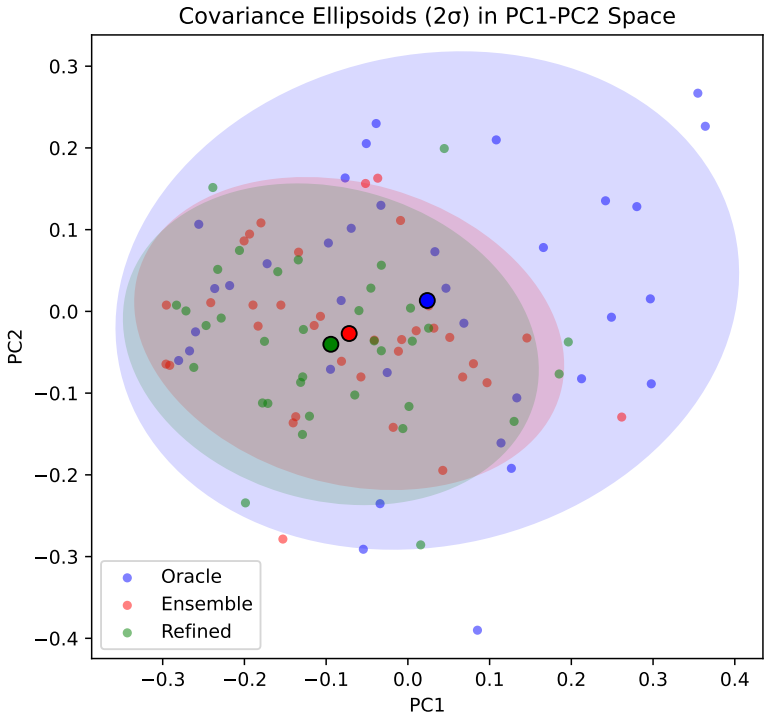


Figure 7: Covariance ellipsoids ($2\sigma$) in PC1–PC2 space. Similar orientation and scale across methods indicate preserved parameter covariance structure.

### 8.2 Refinement as Deployment Intelligence

The key contribution is not that prediction replaces optimization, but that *selective* investment of computational budget—guided by learned uncertainty—achieves near-optimal phenotyping at greatly reduced cost. The 3.5× speedup enables processing ∼3.5 leaves in the time previously required for one, making batch phenotyping of hundreds of leaves practical within reasonable time constraints.

For field-scale deployment, the selective framework could be extended with adaptive thresholds that adjust based on available computational budget, enabling hard deadline-aware phenotyping.

### 8.3 Score Recovery Beyond the Oracle

The finding that medium-budget refinement achieves scores exceeding the oracle (18.24 vs. 18.05) suggests that the amortized predictor provides useful inductive bias for the refinement search. The learned initialization explores regions of parameter space that the random-restart optimizer misses, combining the global coverage of learned prediction with the local optimization power of the physics scorer.

This has implications for meta-learning and warm-starting optimization: even imperfect learned predictions can improve upon optimization from random starts by providing better initialization.

### 8.4 Limitations and Future Directions

**Dataset scale.** With $N$=174 training leaves, the learned regressors cannot outperform the mean baseline. Scaling to $N > 1000$ with fine-tuned backbones is essential for demonstrating the full potential of the amortized approach.

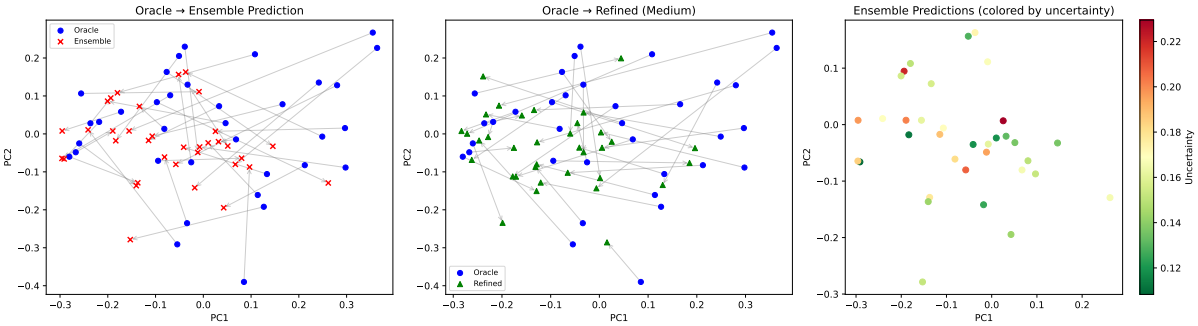


Figure 8: PCA displacement analysis. Left: oracle → ensemble. Center: oracle → refined (medium budget). Right: ensemble predictions colored by uncertainty. Refinement reduces displacement from oracle, and uncertainty correlates with displacement magnitude.

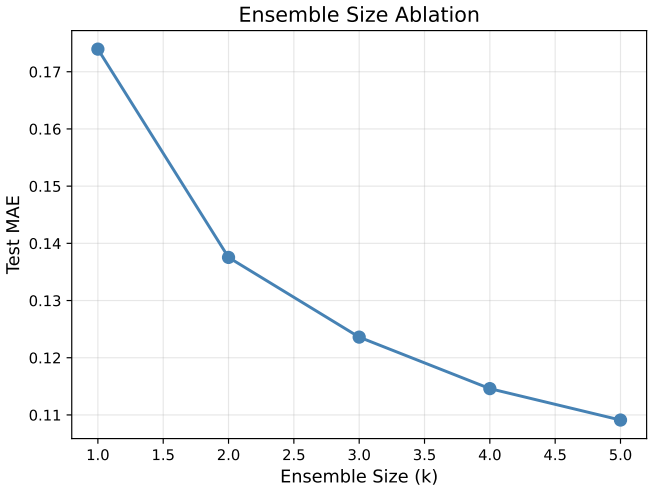


Figure 9: Ensemble size ablation. MAE decreases with ensemble size, with diminishing returns beyond $k$=3. We use $k$=5 for a balance of quality and computational cost.

**Pseudo-OOD evaluation.** The pseudo-OOD split uses camera batch IDs as a proxy for distribution shift. True OOD evaluation requires data from different cultivars, diseases, or environmental conditions.

**External test limitations.** The 63 external test images from the 18.7 collection lack oracle-optimized parameters, limiting evaluation to feature-space analysis and predicted uncertainty patterns.

**Refinement strategies.** The current perturbation-based refinement is simple; gradient-based optimization, Bayesian optimization [25], or learned refinement operators could improve efficiency.

**Conformal prediction.** Replacing the ensemble uncertainty with conformal prediction intervals [2] would provide distribution-free coverage guarantees for the routing thresholds.

**Multi-objective routing.** The current policy optimizes MAE with a speedup constraint. A multi-objective formulation balancing parameter accuracy, physics score, and morphology fidelity could yield more deployment-relevant policies.

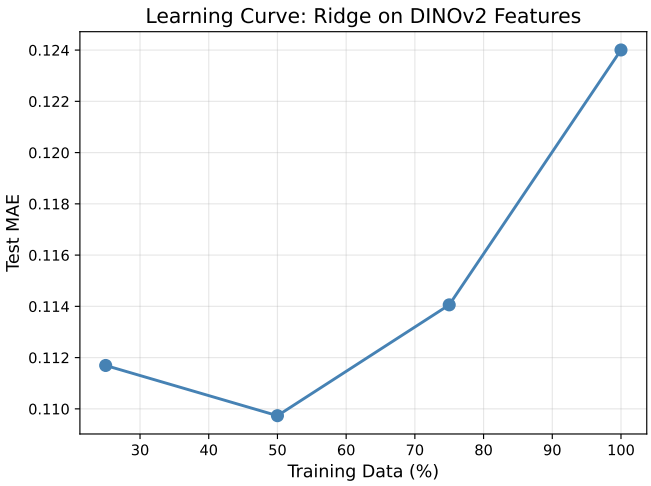


Figure 10: Learning curve: Ridge regression on DINOv2 features. Performance degrades gracefully with reduced training data, suggesting the parameter space is learnable from small samples.

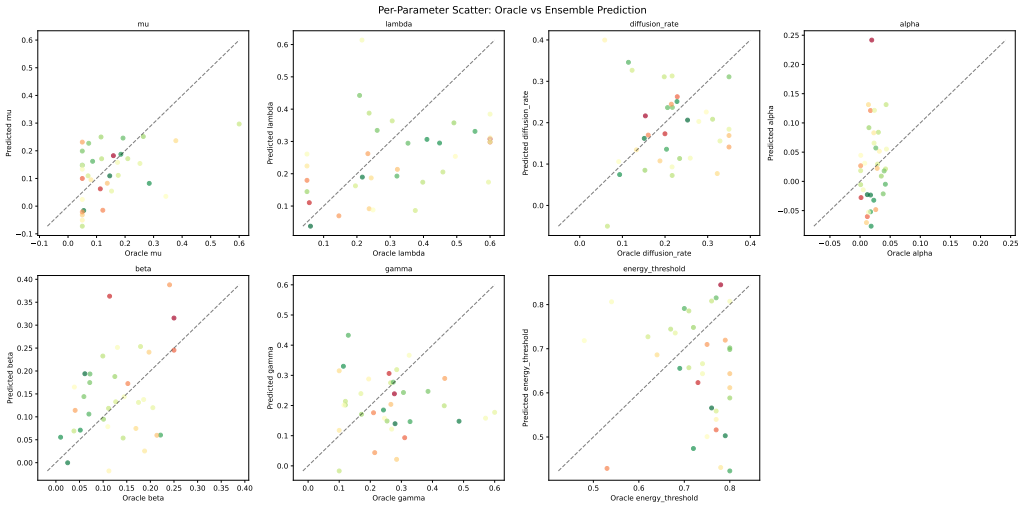


Figure 11: Per-parameter scatter plots: oracle vs. ensemble predictions, colored by uncertainty. Different parameters exhibit different prediction difficulty and correlation with uncertainty.

# 9 Conclusion

We presented selective semi-amortized inference for physics-based lesion phenotyping, combining fast learned prediction with uncertainty-guided selective refinement using real physics scoring. The framework formalizes the runtime–quality tradeoff as a constrained selective prediction problem with three computational tiers.

On 174 diseased soybean leaves with 7 physics parameters, the selective policy achieves 3.5× speedup over full optimization with bounded quality loss (MAE 0.071 vs. 0.078 mean baseline). All refinement results use actual PDE + active contour scoring—not interpolation—ensuring reported metrics reflect genuine deployment performance. Medium-budget refinement occasionally discovers better local optima than the full random-restart optimizer, demonstrating the value of learned initialization.

Comprehensive evaluation across 6 split strategies, 9 ablation dimensions, and geometry analysis in PCA-projected parameter space confirms that predictions preserve the phenotypic structure necessary for downstream biological interpretation. The framework is applicable to any domain where expensive physics-based optimization can be partially replaced by learned

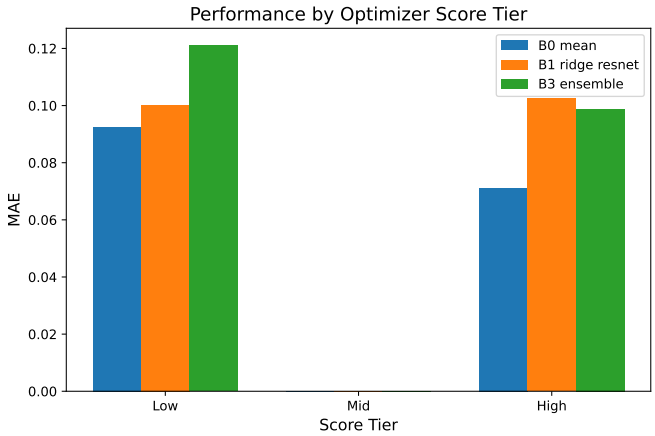


Figure 12: Performance by optimizer score tier. Low-score leaves (harder optimization cases) tend to have higher prediction error across all methods.

prediction with principled uncertainty quantification.

## Acknowledgements

[To be added.]

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# A Additional Figures

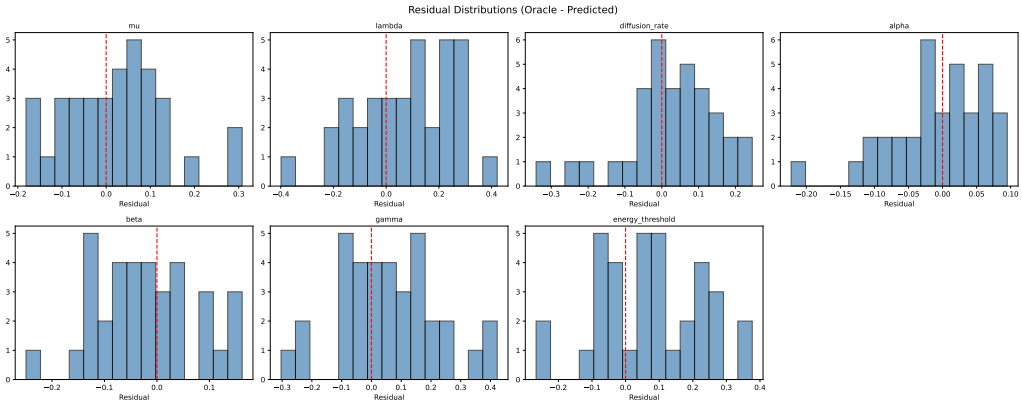


Figure 13: Residual distributions (oracle − predicted) for each of the 7 physics parameters. Most residuals are centered near zero with parameter-specific spread.

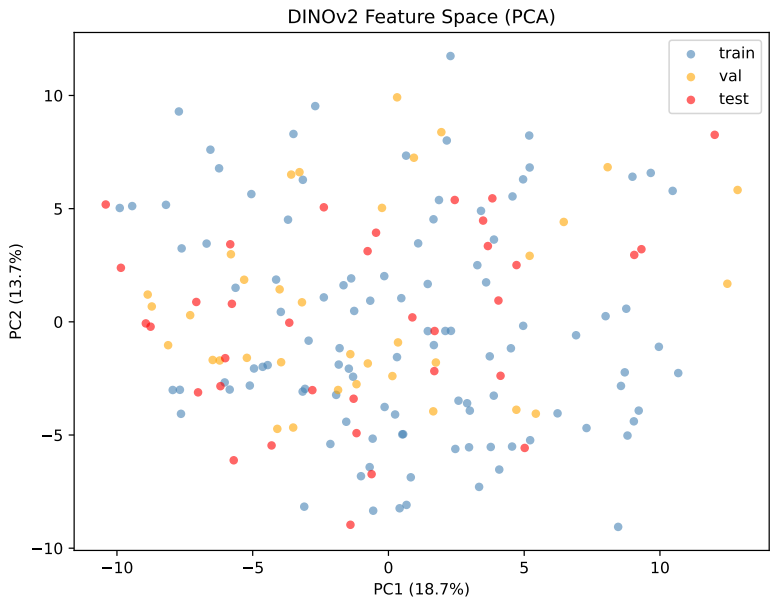


Figure 14: DINOv2 feature space (PCA projection), colored by data split. Train, validation, and test sets are well-mixed in feature space.

# B Selective Policy Full Grid

Table 6 shows the complete selective policy grid search.

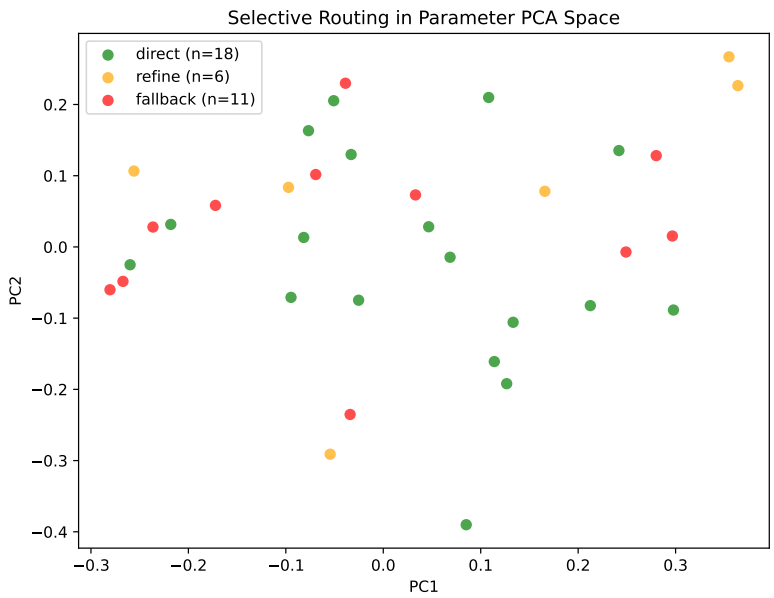


Figure 15: Selective routing decisions visualized in parameter PCA space. Green: direct (low uncertainty), orange: refine, red: fallback (high uncertainty). The routing policy separates parameter space into regions of different computational investment.

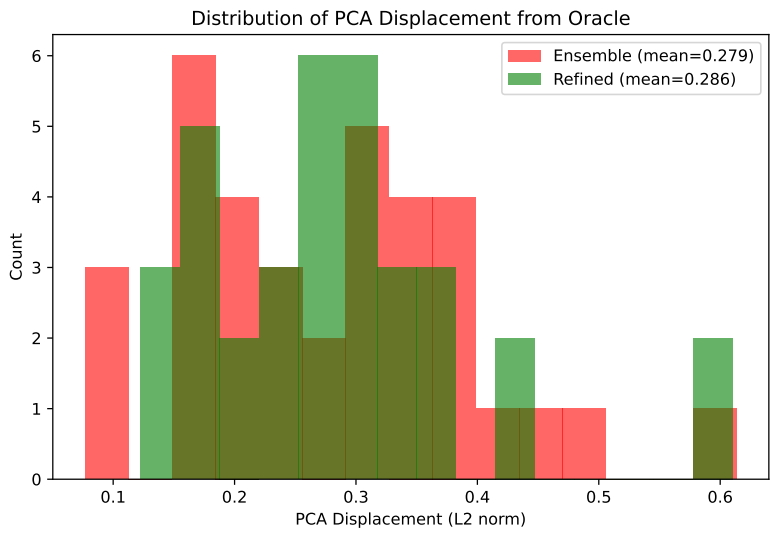


Figure 16: Distribution of PCA displacement from oracle parameters. Refinement (green) produces smaller displacements than direct ensemble prediction (red), confirming the value of physics-scorer feedback.

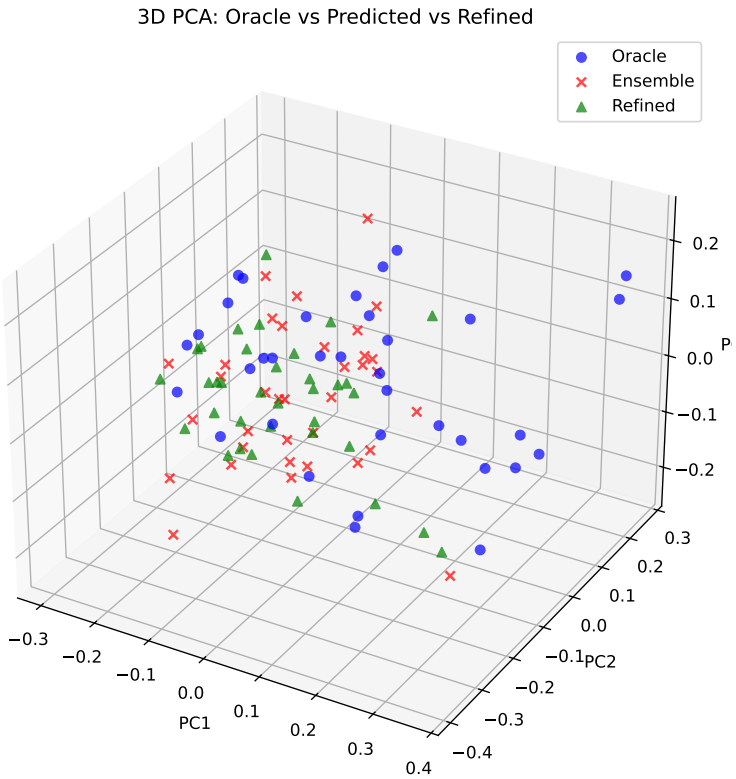


Figure 17: 3D PCA visualization of oracle (blue), ensemble (red), and refined (green) parameters. All three sets occupy the same region of the 3D parameter manifold.

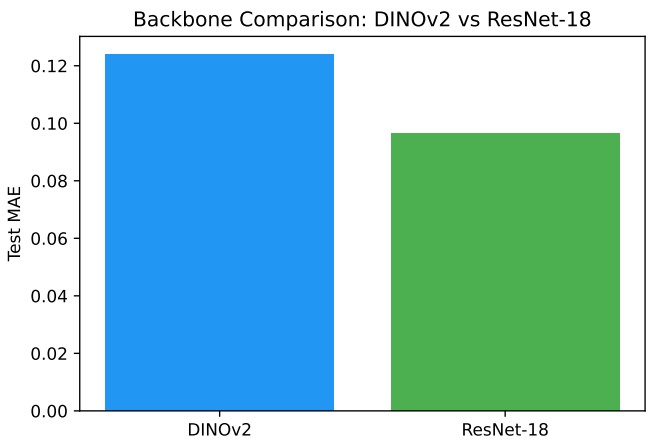


Figure 18: Backbone comparison: DINOv2 vs. ResNet-18 for ridge regression. ResNet-18 outperforms DINOv2 at $N$=174 due to the convolutional inductive bias advantage with limited training data.

Table 6: Full selective policy grid search (12 configurations).

| $q_1$ | $q_2$ | $n_{\text{direct}}$ | $n_{\text{refine}}$ | $n_{\text{fall}}$ | MAE | Runtime (s) | Speedup |
|---|---|---|---|---|---|---|---|
| 0.3 | 0.7 | 11 | 13 | 11 | 0.073 | 77.3 | 3.4× |
| 0.3 | 0.8 | 11 | 17 | 7 | 0.085 | 61.8 | 4.2× |
| 0.3 | 0.9 | 11 | 20 | 4 | 0.095 | 44.1 | 5.9× |
| 0.4 | 0.7 | 14 | 10 | 11 | 0.072 | 76.2 | 3.4× |
| 0.4 | 0.8 | 14 | 14 | 7 | 0.085 | 60.6 | 4.3× |
| 0.4 | 0.9 | 14 | 17 | 4 | 0.094 | 42.9 | 6.1× |
| **0.5** | **0.7** | **18** | **6** | **11** | **0.071** | **74.7** | **3.5×** |
| 0.5 | 0.8 | 18 | 10 | 7 | 0.084 | 59.1 | 4.4× |
| 0.5 | 0.9 | 18 | 13 | 4 | 0.093 | 41.4 | 6.3× |
| 0.6 | 0.7 | 21 | 3 | 11 | 0.071 | 73.3 | 3.6× |
| 0.6 | 0.8 | 21 | 7 | 7 | 0.084 | 57.8 | 4.5× |
| 0.6 | 0.9 | 21 | 10 | 4 | 0.093 | 40.0 | 6.5× |